07. Model Training & Hyperparameter Tuning

Document what the report says was trained and what the notebook merely loads.

Project: Automated Mobile-Technology Brand Sentiment and Headline Generator

Source archive: NLP-Sentiment-Classification-of-Mobile-brands-main.zip | Notebook: main.ipynb | Supporting report and presentation included

Theme classifiers

The report specifies separate TF-IDF + XGBoost classifiers for articles and tweets. The original train/validation/test split is 3,000/600/400 per type. The exact TF-IDF parameters, XGBoost hyperparameters and training code are not present in the supplied archive, so they cannot be reproduced or quoted accurately.

Sentiment model development

- The team hand-labelled around 200 tweets for model selection and evaluation, according to the presentation.
- An external mobile-product review dataset was curated by combining random pairs of texts; it did not transfer well to the challenge validation sample.
- The team then fine-tuned BERT checkpoints trained on Hinglish and Indian-domain text, reporting satisfactory validation behaviour qualitatively.
- Multiple sentiment methods were explored: text-pair BERT, local contextual embeddings, dependency-tree features, lexicon weighting and TF-IDF single-brand classifiers.
- Long articles are split into chunks and chunk-level outputs combined.

T5 headline model

The report says T5-base was fine-tuned for two epochs on translated English article data, with early stopping based on validation ROUGE and BLEU. PEGASUS and Lead-3 were compared and dropped because they did not outperform T5-base on the selected metric. Exact final metric values and full training scripts are not included.

Inference notebook sequence

- Downloads tweet/article TF-IDF vectorizers and classifiers, T5 headline checkpoint, multilingual BERT sentiment checkpoint and additional regression/TF-IDF artifacts from Google Drive.
- Loads evaluation_data.xlsx and splits it into article and tweet subsets.
- Cleans text, extracts brands, predicts mobile-tech theme, detects language, translates non-English mobile-tech records.
- Generates headlines for mobile-tech articles, segments brand contexts and predicts per-brand sentiment.
- Exports headline-output.csv and sentiment-output.csv.

Tuning status and reproducibility

The report describes validation-based model selection and early stopping for T5, but the supplied notebook does not show systematic hyperparameter search. Exact XGBoost parameters, BERT fine-tuning settings, sentiment ensemble weights and T5 ROUGE/BLEU values are absent. Avoid inventing them in an interview.